

Queries Specification Document

Team # 5

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Topic: Personal Finance Management

This document describes the data operations and reporting queries for the Personal Finance Management System. It covers user and account management, transactions and transfers, bills, budgets, loans, and goals, as well as analytics and scheduled tasks.

No SQL is included, these are the descriptions to be implemented in Phase Four.

Basic Operations

User Profile

Register a new user in the system

Create a user profile with full name, email and password (hashed by the app). Enforce unique email, require password, initialize defaults (currency, starter categories).

Update user profile information

Allow changes to full name, email (ensure uniqueness), subscription status, and password via secure reset flow.

View profile details

Return current user attributes (full name, email, subscription, created_at, status).

Deactivate user

Set status to 'inactive'; block sign-in and money-impacting actions while retaining data for audit.

Delete user

Hard-delete the user with all its data.

User Accounts

Open a financial account

Create an account with name, number, type (cash/credit card/debit card/current account/investment), currency, initial balance, and allow-overdraft flag.

Update financial account information

Permit updates to account name, number, type and allow-overdraft flag.

Select accounts

List all the accounts of the user, including the current balance and created date.

Close a financial account

Allowed only when balance is zero, otherwise prompt to transfer or settle.

Deactivate the account and record the closed date.

Delete a financial account

Delete only if no transactions exist, otherwise deactivate the account.

Categories**Create new category**

Provide name, type (income/expense), and optional parent category id. Prevent cycles in the hierarchy.

Select categories list (by parent)

Retrieve child categories for a given parent.. If no parent is given, return the parent categories. Optionally, return the full tree.

Update category information

Rename or re-parent the given category.

Delete an existing category

Allowed only if no transactions are recorded by the given category.

Budget**Create budget**

Provide category, limit amount, period type (monthly/weekly/custom), and alert threshold flag. Reject overlapping duplicate budgets.

Update budget

Allow changes to limit amount, period type, alert threshold and is_active flags.

See budgets list

List user's all budgets.

Delete budget

Remove or archive a budget that has not yet started, otherwise mark inactive.

Bill

Create bill

Provide bill name, amount, recurring frequency, due date, and notes.

Update bill information

Edit bill name, amount, recurring frequency, due date, notes.

See bills list

List unpaid, upcoming, and overdue bills with days remaining/overdue and recurrence info.

Delete bill

Allowed only if no payment transaction is linked; otherwise archive.

Loan

Create loan

Provide loan name, principal amount, remaining amount, payment amount, next due date, interest rate %, start date, end date (optional).

Update loan information

Adjust payment amount, interest rate, next due date, or status.

See loans list

List active/closed loans with remaining amount, next due date, and days past due.

Delete loan

Only if no transactions were ever posted to the loan; otherwise mark archived.

Goal

Set a goal

Provide goal name, target amount, initial saved amount, deadline (optional), and notes.

Update goal information

Edit name, target amount, saved amount, deadline, notes, or status.

See goals list

Show progress (saved/target), percent complete, and estimated time to reach based on recent contributions.

Delete goal

Allowed when no linked contribution transactions exist; otherwise archive.

Main / Functional Operations

Transactions

Post a transaction

Record account, transaction type (income/expense/transfer), amount, timestamp, category, and description. Adjust the account balance and enforce an idempotency key (operation_id) to prevent duplicates.

Expense side-effects

Optionally link to a bill, loan, or goal to the transaction:

Bill: set is_paid=true for the period.

Loan: decrease remaining_amount by payment; set is_paid=true; when remaining_amount reaches 0, set status='closed'.

Goal: increase saved_amount; when saved_amount $\geq$ target_amount, set status='achieved'.

Transfer transactions

Require 'from' and 'to' accounts; record two linked legs (debit from source, credit to destination). For cross-currency transfers, compute the target amount using the effective exchange rate on the timestamp.

Change transaction

Allow edits to account, type, amount, timestamp, category, description, and links (bill/loan/goal). Propagate adjustments to balances and linked entities with a full audit trail.

Delete transaction

Reverse effects on balances and linked entities.

Scheduled Operations

Notify about upcoming transactions

For bills: consider recurrence and is_paid.

For loans: consider next_due_date and is_paid. Generate reminders 1 day in advance and on the due date.

Update is_paid status

Bills: when a new period starts, reset is_paid=false.

Loans: when next_due_date passes and a payment was recorded, set is_paid=false and advance next_due_date per schedule.

User-Focused Queries

Account & Transaction Queries

View account history and current balance

Show recent transactions with running balance; filter by date range, amount range, and category.

Check upcoming scheduled payments for an account

List unpaid bills and loan payments within a selected date range, with due dates and projected post-payment balances.

Review monthly statements

Generate period statements with opening balance, total income, total expenses, transfers and closing balance.

Category statement for a period

Summarize income/expense by categories (respecting parent/child hierarchy) with totals and percentages.

Currency and Transfer Queries

See all logs of a transfer between accounts

Display the paired transactions (debit and credit) including currency conversion and timestamps; link by transfer_id.

Track current and historical exchange rates

Show latest rates per currency and the effective rate used for a given transaction date/time, with source and updated_at.

Get all transactions involving a specific category or bill/loan/goal

Filter by category (and descendants) or by a specific bill/loan/goal; include counts and totals.

Reporting Queries

Cash Flow and Financial Reports

Monthly cash-flow report

Group income and expenses by month; compute totals and net amount with trend vs previous months.

Income by category

Identify which income categories contribute most to total income over a selected period.

Largest expenses (descending)

Rank the biggest expense transactions in the selected period and show their categories.

Balance trend for a period

Return daily end-of-day balances across the specified period for charting and forecasting.

Budget performance summary

For each active budget, show limit, actual spend, utilization %, and status (OK/Warn/Exceeded).

User Behavior & System Trends

Recurring subscription detection

Detect the categories and descriptions that repeat at regular intervals with similar amounts to surface subscriptions for review/cancellation.

Account activity comparison

Compare number of transactions and average ticket size per account to identify most/least used accounts within a timeframe for review.